

Patient-Aware Machine-Learning Classification of Benign and Malignant Breast Lesions Using Single- and Multi-Source MRI Radiomic Features

Abstract

Background: Radiomic features from multiple MRI sources may improve classification, but combining more sources can also increase model complexity.

Methods: Five MRI-derived feature sources (ADC, Pre, Post1, Post2, and T2) were evaluated in all 31 possible combinations. Patient-aware nested cross-validation was used. Three feature-selection strategies, optional SMOTE, and ten classifiers were compared. Mean outer-fold ROC-AUC was the primary evaluation measure.

Results: The best pipeline used ADC, Post1, and Pre features with Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM. It achieved a mean ROC-AUC of 0.836 ± 0.088 and a pooled ROC-AUC of 0.799. The pooled PR-AUC, balanced accuracy, and F1-score were 0.832, 0.713, and 0.746, respectively. Combining all five sources produced a lower mean ROC-AUC of 0.747 ± 0.117 . Elastic Net followed by mRMR was the best feature-selection strategy in 20 of 31 configurations, while the overall effect of SMOTE was small.

Conclusion: Selective multi-source integration performed better than complete feature concatenation in this cohort. Larger independent datasets are needed to confirm the performance of the selected ADC–Post1–Pre pipeline.

1 Introduction

Machine learning can identify useful patterns in radiomic data and use them to classify lesions. In radiomics, each lesion is represented by many quantitative features describing properties such as intensity, texture, and shape.

Developing reliable models from radiomic data is challenging because the number of features is often large compared with the number of patients. This increases the risk of overfitting. Missing values, correlated features, different numerical scales, class imbalance, and model complexity may also affect performance. Careful preprocessing, feature selection, and validation are therefore required.

Radiomic features from different MRI sequences can be analysed separately or combined. Although combining sources may provide more information, it also increases the number of fea-

tures without increasing the number of patients. Therefore, using more imaging sources does not necessarily improve classification performance.

Another important issue is patient-level data leakage. When a patient has more than one region of interest, all regions from that patient must remain in the same cross-validation fold. Otherwise, related samples may appear in both the training and test sets and produce overly optimistic results.

In this study, we developed a patient-aware machine-learning framework using five sets of pre-extracted MRI radiomic features: ADC, Pre, Post1, Post2, and T2. All 31 non-empty source combinations were evaluated, including single-source and multi-source configurations. For each configuration, three feature-selection strategies, SMOTE and no-SMOTE conditions, and ten classifiers were compared using patient-aware nested cross-validation.

The main objective was to identify the combi-

nation of imaging sources, feature-selection strategy, class-balancing condition, and classifier that achieved the best performance for distinguishing benign and malignant breast lesions. We also examined the effect of combining additional imaging sources, the usefulness of SMOTE, and the stability of selected features across patient groups.

2 Materials and Methods

2.1 Computational Study Design

A supervised machine-learning framework was developed to classify breast lesions as benign or malignant using pre-extracted radiomic features. The analysis started from tabular datasets and did not include image segmentation, image preprocessing, or feature extraction.

Five MRI feature sources were evaluated: apparent diffusion coefficient (ADC), pre-contrast imaging (Pre), two post-contrast phases (Post1 and Post2), and T2-weighted imaging (T2). Each source was analysed separately and in combination with the other sources. Multi-source datasets were created by joining the feature columns of matched regions of interest (ROIs).

Each row represented one ROI with a benign or malignant label. Because some patients contributed more than one ROI, the ROI was the prediction unit, while the patient was the grouping unit during validation.

All non-empty combinations of the five sources were examined:

$$\binom{5}{1} + \binom{5}{2} + \binom{5}{3} + \binom{5}{4} + \binom{5}{5} = 5 + 10 + 10 + 5 + 1 = 31. \quad (1)$$

These configurations included five single-source, ten pairwise, ten triple-source, five quadruple-source, and one five-source dataset.

The same workflow was applied to all 31 configurations. It included data-quality checks, training-fold preprocessing, feature selection, automatic selection of the number of features, optional SMOTE, hyperparameter optimisation, model fitting, and performance evaluation.

An overview of the study design is shown in Figure 1.

Three feature-selection strategies were tested with and without SMOTE and combined with ten classifiers. Each dataset configuration therefore included

$$3 \times 2 \times 10 = 60 \quad (2)$$

candidate pipelines, giving 1,860 primary pipeline configurations across the complete study.

To prevent data leakage, all ROIs from the same patient remained in the same cross-validation partition. Imputation, variance filtering, standardisation, feature selection, SMOTE, and hyperparameter tuning were performed only within the relevant training data. Outer-test data were used only for final evaluation.

2.2 Input Datasets and Prediction Task

The analysis used comma-separated value (CSV) files containing pre-extracted numerical radiomic features. The original MRI images were not provided to the machine-learning models. Instead, each row represented one region of interest (ROI), and each numerical column represented one candidate predictor.

Each row also contained a patient identifier, an ROI identifier, and a binary class label. The patient identifier was stored in `INFO_PatientName` and renamed to `PatientID`. The ROI identifier was stored in `INFO_NameOfRoi`, and the target label was stored in `Label`. These columns were used for data matching, quality control, and patient-aware splitting, but were excluded from the predictor matrix.

The target labels were encoded as

$$y_i = \begin{cases} 0, & \text{benign,} \\ 1, & \text{malignant.} \end{cases} \quad (3)$$

For a dataset with n ROIs and p predictor features, the input data were represented by

$$\mathbf{X} \in \mathbb{R}^{n \times p}, \quad \mathbf{y} \in \{0, 1\}^n, \quad \mathbf{g} \in \mathbb{N}^n, \quad (4)$$

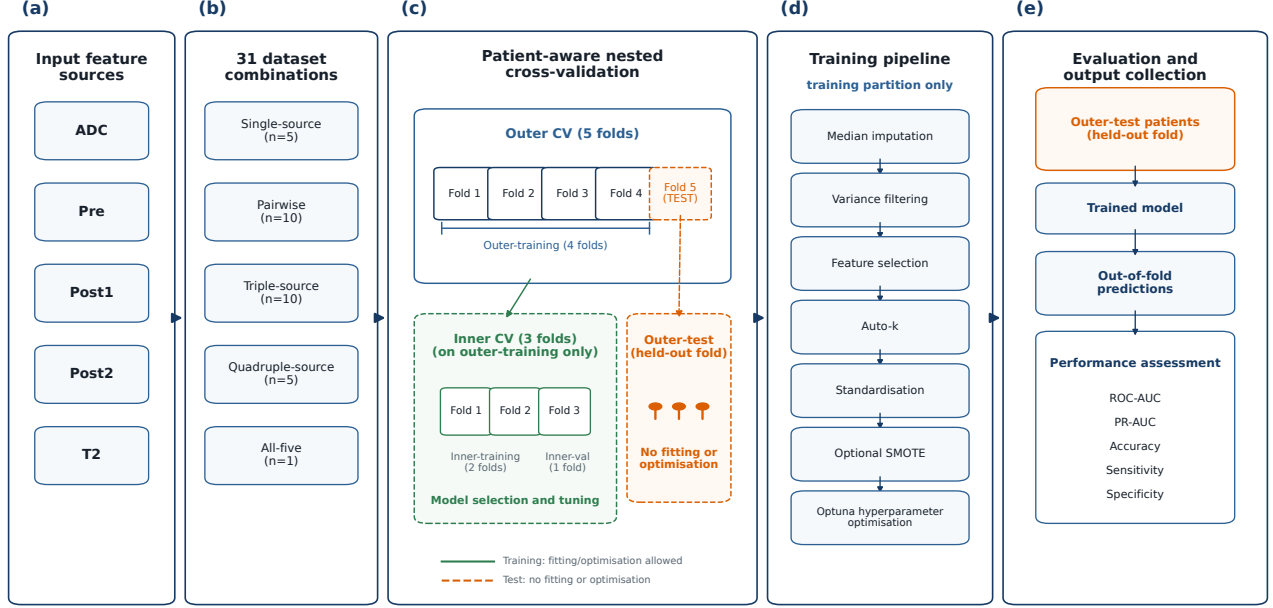


Figure 1: **Overview of the computational workflow.** Five radiomic feature sources were evaluated in 31 single- and multi-source configurations. Models were developed using patient-aware nested cross-validation. Imputation, variance filtering, feature selection, standardisation, optional SMOTE, and hyperparameter optimisation were performed using training data only. Final performance was measured on the outer-test patients. ADC, apparent diffusion coefficient; ROI, region of interest; SMOTE, Synthetic Minority Over-sampling Technique.

where $\mathbf{X}$ is the feature matrix, $\mathbf{y}$ contains the class labels, and $\mathbf{g}$ contains the patient identifiers.

Predictions were generated for individual ROIs. However, the patient was used as the grouping unit during cross-validation because some patients contributed more than one ROI. All ROIs from the same patient were therefore kept in the same training, validation, or test partition. This prevented patient-level data leakage.

Five single-source datasets were available: ADC, Pre, Post1, Post2, and T2. Together, they represented 48 unique patients. The datasets contained between 114 and 117 ROIs and between 144 and 145 predictor features, as summarised in Table 1.

The class imbalance was limited. Therefore, SMOTE was not applied automatically. Models with and without SMOTE were evaluated as separate experimental conditions, as described in Section 2.8.

2.3 Construction of Multi-Source Datasets

Multi-source datasets were created by joining the feature columns of matching regions of interest (ROIs) from different MRI sources. The datasets were concatenated column-wise, so combining sources increased the number of predictor features but did not create new patients or ROIs.

For a set of sources $\mathcal{S} = \{s_1, \dots, s_m\}$, the combined feature matrix was defined as

$$\mathbf{X}^{(\mathcal{S})} = [\mathbf{X}_{\mathcal{I}_{\mathcal{S}}}^{(s_1)} \mid \dots \mid \mathbf{X}_{\mathcal{I}_{\mathcal{S}}}^{(s_m)}], \quad (5)$$

where $\mathcal{I}_{\mathcal{S}}$ is the set of ROIs available in all selected sources. Only shared ROIs were retained, so the number of samples could differ slightly between source combinations.

Before matching, empty columns and exported index columns beginning with `Unnamed:` were re-

Table 1: Characteristics of the five single-source datasets. Class 0 represents benign ROIs, and class 1 represents malignant ROIs. Metadata and identifier columns were excluded from the feature counts.

Feature source	ROIs, n	Predictor features, n	Class 0, n	Class 1, n
ADC	114	144	50	64
Pre	117	145	52	65
Post1	116	145	52	64
Post2	117	145	52	65
T2	117	145	52	65

moved. Patient and target columns were standardised to PatientID and Label.

ROIs were primarily matched using INFO_NameOfRoi. When necessary, a combined key containing the patient and ROI identifiers was used. A key was accepted only when it was unique within every selected source, preventing unintended many-to-many joins. Row-order matching was allowed only when the datasets had the same number of rows and an identical sequence of patient identifiers.

The datasets were combined using a one-to-one inner join. Therefore, an ROI was retained only when it was present in every source included in the combination. Dataset construction was stopped if the matching key was duplicated or no matching observations were found.

Because the original feature names were often repeated across sources, each predictor was prefixed with its source name. For example,

ADC__GLCM_Correlation
Post1__GLCM_Correlation
T2__GLCM_Correlation

were treated as separate features. Shared metadata columns were not given source-specific prefixes.

The patient identifier and class label were also checked for agreement across the matched sources:

$$y_i^{(s_1)} = \dots = y_i^{(s_m)}, \quad g_i^{(s_1)} = \dots = g_i^{(s_m)}. \quad (6)$$

If a mismatch was detected, it was recorded and construction of that dataset was stopped. After

validation, only one PatientID column and one Label column were retained.

The same procedure was used to construct all pairwise, triple-source, quadruple-source, and five-source datasets. A creation log recorded the selected sources, matching method, retained ROIs, feature counts, class distribution, and any detected errors.

2.4 Data Preparation and Quality Control

Before model development, all datasets underwent the same quality-control procedure. This stage checked the dataset structure, identified invalid or duplicated records, and defined the numerical predictor features. No model fitting was performed.

Each dataset was required to contain a patient identifier, an ROI identifier, and a binary target label. The INFO_PatientName column was renamed to PatientID, while INFO_NameOfRoi and Label were retained. The target was required to contain only 0 and 1. Identifier, label, exported index, and other non-predictive metadata columns were excluded from the feature matrix.

All remaining numerical columns were treated as candidate predictors. Columns containing no valid numerical values were removed. Rows with partially missing feature values were retained because imputation was later performed within the cross-validation training folds.

Feature names were cleaned by removing spaces, parenthetical text, and unsupported characters. Common feature-family names were shortened to Morph, IB, and IH. Numerical suffixes

were added when cleaning produced duplicated names, and the mapping between original and cleaned names was saved.

Missing values, infinite values, and feature variance were recorded. Features with variance less than or equal to 10^{-12} were marked as constant, while features with variance up to 10^{-8} were marked as near-zero variance. These checks were descriptive; variance filtering and median imputation were fitted again within each training partition.

The workflow also checked for duplicated rows, repeated ROI identifiers, duplicated patient–ROI pairs, and repeated feature names. Possible duplicates were reported but were not removed automatically.

Potential outliers were identified using the modified z -score,

$$z_{ij}^* = 0.6745 \frac{x_{ij} - \text{median}(x_j)}{\text{MAD}(x_j)}. \quad (7)$$

Values with $|z_{ij}^*| > 3.5$ were flagged. Outlier detection was used only for quality control; no ROI was removed, and no global clipping or winsorisation was applied.

Patient-level ROI counts and class distributions were also recorded. Patients with multiple ROIs or ROIs from both classes were retained, but all ROIs from the same patient were later assigned to the same cross-validation partition.

No global imputation, standardisation, feature selection, SMOTE, hyperparameter optimisation, or classifier training was performed during quality control. These operations were applied only to the relevant training data during patient-aware nested cross-validation.

2.5 Patient-Aware Nested Cross-Validation

Model development and evaluation were performed using patient-aware nested cross-validation. The outer loop estimated performance on unseen patients, while the inner loop was used to select the number of features and optimise classifier hyperparameters.

2.5.1 Patient-aware data splitting

Because some patients contributed more than one region of interest (ROI), the patient identifier was used as the grouping variable. All ROIs from the same patient were assigned to the same fold. For outer fold q ,

$$\mathcal{G}_{\text{train}}^{(q)} \cap \mathcal{G}_{\text{test}}^{(q)} = \emptyset, \quad (8)$$

where $\mathcal{G}_{\text{train}}^{(q)}$ and $\mathcal{G}_{\text{test}}^{(q)}$ are the training and test patient groups.

The absence of patient overlap was checked programmatically in every fold. Splits were generated using `StratifiedGroupKFold`, which kept patient groups together while preserving the class distribution as closely as possible. The data were shuffled using a fixed random seed of 100.

2.5.2 Outer cross-validation loop

The outer loop contained five patient-aware folds:

$$K_{\text{outer}} = 5. \quad (9)$$

In each fold, approximately four-fifths of the patients were used for model development, and the remaining patients formed the outer-test set. The outer-test data were not used for imputation, variance filtering, feature selection, standardisation, SMOTE, or hyperparameter optimisation.

After all modelling decisions were completed, the selected pipeline was fitted to the complete outer-training data and evaluated once on the outer-test data.

2.5.3 Training-only preprocessing

Preprocessing parameters were estimated from the outer-training data. Missing values in both the training and test sets were replaced using training-set medians. A variance filter was then fitted to the outer-training data, and features with variance less than or equal to 10^{-10} were removed.

After feature selection, the retained features were standardised as

$$z_{ij} = \frac{x_{ij} - \mu_{j,\text{train}}}{\sigma_{j,\text{train}}}, \quad (10)$$

where the mean and standard deviation were calculated from the training data and applied unchanged to the corresponding validation or test data.

2.5.4 Inner cross-validation loop

The outer-training data were further divided using three patient-aware inner folds:

$$K_{\text{inner}} = 3. \quad (11)$$

The inner loop compared candidate feature counts and classifier hyperparameters using ROC-AUC. Patient groups remained separate between the inner-training and inner-validation sets. When three valid folds were not possible, the number of inner folds was reduced to the largest feasible value, with at least two folds required.

For automatic feature-count selection, the inner folds were generated after outer-fold median imputation and variance filtering. Within each inner split, feature selection and standardisation were fitted using the inner-training data and applied to the inner-validation data. When enabled, SMOTE was applied only to the transformed inner-training samples.

The best inner-loop settings were then used to fit the model on the complete outer-training partition.

2.5.5 Outer-fold evaluation and out-of-fold predictions

The final model generated continuous prediction scores and class labels for the outer-test ROIs. Each ROI appeared in an outer-test fold once, and its prediction was produced without using that ROI or another ROI from the same patient during training.

Predictions from all outer folds were combined into the out-of-fold prediction vector

$$\hat{\mathbf{p}}_{\text{OOF}} = [\hat{p}_1, \hat{p}_2, \dots, \hat{p}_n]^T. \quad (12)$$

These predictions were used to calculate pooled performance metrics and construct ROC curves, precision–recall curves, confusion matrices, and calibration summaries. Fold-level results were

also retained to measure variation across different test-patient groups.

2.6 Feature-Selection Strategies

Feature selection was used to reduce the number of predictors before classifier training. This was especially important for multi-source datasets, where the number of features increased without a similar increase in the number of patients.

Three sequential strategies were compared:

1. correlation filtering followed by mRMR;
2. Elastic Net followed by mRMR;
3. mRMR followed by Elastic Net.

The order of the methods was treated as part of the pipeline because the first method determined which features were available to the second. All feature-selection steps were fitted using training data only. The selected feature indices were then applied unchanged to the corresponding validation or test data. SMOTE was applied only after feature selection and standardisation.

2.6.1 Correlation Filtering

Correlation filtering removed strongly redundant predictors. Two features were considered highly correlated when

$$|r_{jk}| > 0.90, \quad (13)$$

where r_{jk} is their Pearson correlation.

To decide which feature to retain from a correlated group, each feature was ranked using a direction-independent univariate ROC-AUC:

$$A_j^* = \max(A_j, 1 - A_j). \quad (14)$$

Features were processed from highest to lowest A_j^* . After a feature was retained, other features correlated with it above the threshold were removed. Correlations and ROC-AUC values were calculated using training data only.

2.6.2 Minimum Redundancy Maximum Relevance

Minimum redundancy maximum relevance (mRMR) selected features that were informative about the target while limiting repeated information.

Feature relevance was estimated using mutual information:

$$R_j = I(X_j; Y), \quad (15)$$

where X_j is feature j and Y is the class label. Mutual information was calculated using `mutual_info_classif`.

Redundancy was defined as the mean absolute correlation between a candidate feature and the already selected features:

$$D_j = \frac{1}{|\mathcal{F}|} \sum_{s \in \mathcal{F}} |r_{js}|, \quad (16)$$

where $\mathcal{F}$ is the current selected set. Candidate features were ranked using

$$Q_j = R_j - D_j. \quad (17)$$

The first feature was selected using relevance alone. Additional features were added one at a time until the required feature count was reached.

2.6.3 Elastic-Net Feature Selection

Elastic-Net selection was implemented using regularised logistic regression. It combines L_1 regularisation, which can reduce some coefficients to zero, with L_2 regularisation, which improves stability among correlated predictors.

The selector used an `l1_ratio` of 0.8, $C = 0.1$, the `saga` solver, balanced class weights, a maximum of 5000 iterations, a tolerance of 10^{-5} , and a random seed of 100. Features were standardised before fitting and ranked by the absolute value of their coefficients:

$$W_j = |\beta_j|. \quad (18)$$

Both positive and negative coefficients were therefore considered potentially informative.

2.6.4 Sequential Feature-Selection Pipelines

In the first strategy, correlation filtering reduced strongly correlated predictors, and mRMR selected the final subset:

$$\text{Correlation filtering} \rightarrow \text{mRMR} \rightarrow \text{final subset}. \quad (19)$$

In the second strategy, Elastic Net first retained a candidate pool of up to 28 features. mRMR then selected the final subset:

$$\text{Elastic Net} \rightarrow \text{mRMR} \rightarrow \text{final subset}. \quad (20)$$

In the third strategy, mRMR first retained up to 28 candidate features, and Elastic Net ranked and selected the final subset:

$$\text{mRMR} \rightarrow \text{Elastic Net} \rightarrow \text{final subset}. \quad (21)$$

The final number of retained features was determined by the Auto- k procedure described in Section 2.7.

Within each outer fold, the selected strategy was fitted after median imputation and variance filtering. During Auto- k selection, the same procedure was repeated within the inner folds. The final strategy was then refitted using the complete outer-training data and applied unchanged to the outer-test data.

For each dataset, outer fold, feature-selection strategy, and SMOTE condition, the same selected features were supplied to all ten classifiers. Selected feature names were saved for each outer fold and later used to assess feature-selection stability.

2.7 Automatic Selection of the Feature-Set Size

The number of retained features was selected automatically within each outer cross-validation fold. This Auto- k procedure allowed the feature-set size to adapt to the current outer-training data instead of using one fixed value for all datasets.

The candidate feature counts were

$$\mathcal{K} = \{4, 6, 8, 10, 12, 14, 16, 18\}. \quad (22)$$

Values larger than the number of available features after imputation and variance filtering were excluded.

Auto- k was performed separately for each outer fold and feature-selection strategy using only the outer-training data. These data were divided into up to three patient-aware inner folds with `StratifiedGroupKFold`. At least two valid inner folds were required.

For each candidate value k , the active feature-selection pipeline was fitted on the inner-training data and used to select k features. The selected features were standardised using the inner-training mean and standard deviation and then applied to the inner-validation data.

A lightweight Logistic Regression model was used only to compare the candidate feature counts. It used L_2 regularisation, the `liblinear` solver, balanced class weights, a maximum of 5000 iterations, and a random seed of 100.

For each candidate k , the mean inner-validation ROC-AUC was calculated as

$$S_k = \frac{1}{R_k} \sum_{r \in \mathcal{R}_k} \text{AUC}_{k,r}, \quad (23)$$

where $\mathcal{R}_k$ contains the valid inner folds and R_k is their number. Folds containing only one class were excluded from the ROC-AUC calculation.

The selected feature count was

$$k^* = \arg \max_{k \in \mathcal{K}} S_k. \quad (24)$$

When two values produced the same mean ROC-AUC, the smaller feature count was selected.

SMOTE was not applied during Auto- k selection. After k^* was chosen, the feature-selection pipeline was refitted using the complete outer-training data. The resulting feature subset was then applied unchanged to the outer-test data and supplied to all ten classifiers.

The candidate values, inner-fold ROC-AUC scores, and selected feature count were saved for each outer fold to support reproducibility.

2.8 Class-Imbalance Handling Using SMOTE

The numbers of benign and malignant ROIs were slightly different. Because this imbalance was limited, class balancing was evaluated as an optional modelling condition rather than applied automatically.

Each feature-selection strategy was tested under two conditions:

1. without SMOTE;
2. with the Synthetic Minority Over-sampling Technique (SMOTE).

This comparison was repeated for every dataset configuration and classifier.

2.8.1 Position Within the Pipeline

SMOTE generates synthetic minority-class samples using distances between observations. It was therefore applied after feature selection and standardisation:

$$\begin{aligned} \text{Imputation} &\rightarrow \text{variance filtering} \\ &\rightarrow \text{feature selection} \rightarrow \text{standardisation} \\ &\rightarrow \text{optional SMOTE} \rightarrow \text{classifier}. \end{aligned} \quad (25)$$

SMOTE was not used during feature selection or Auto- k selection. This prevented synthetic observations from affecting the selected features or feature count.

2.8.2 Training-Only Application

SMOTE was applied only to training data. During inner-loop hyperparameter optimisation, it was fitted to the transformed inner-training samples after imputation, variance filtering, feature selection, and standardisation. Inner-validation samples were not oversampled.

After hyperparameter selection, SMOTE was applied to the standardised outer-training data before fitting the final model. The outer-test data were transformed using training-derived parameters but retained their original observations and class distribution.

2.8.3 Adaptive Neighbour Selection

The number of minority-class neighbours was adjusted to the size of the current training partition:

$$k_{\text{SMOTE}} = \min(5, n_{\text{minority}} - 1), \quad (26)$$

where n_{minority} is the number of minority-class training samples.

SMOTE was skipped when the training data contained only one class or fewer than two minority-class samples. It was also skipped if re-sampling failed because of a numerical or structural error. A fixed random seed of 100 was used.

2.8.4 Experimental Comparison

The two SMOTE conditions used the same patient-aware folds, feature-selection procedures, candidate feature counts, classifiers, and evaluation metrics. Their results were compared using ROC-AUC, PR-AUC, balanced accuracy, sensitivity, specificity, precision, and F1-score.

2.9 Machine-Learning Classifiers

Ten supervised classifiers were evaluated, covering linear, kernel-based, tree-based, distance-based, and probabilistic methods. Each classifier received the feature subset selected within the corresponding outer fold. Metadata columns, patient identifiers, ROI identifiers, and class labels were excluded from the model input.

All selected features were standardised before model fitting. For ROI i , each classifier produced a continuous positive-class score,

$$\hat{p}_i = f_{\theta}(\mathbf{x}_i), \quad (27)$$

where $\mathbf{x}_i$ contains the selected features and θ represents the fitted model parameters. Predicted probabilities were used when available; otherwise, continuous decision scores were retained. These outputs were used to calculate ROC-AUC and PR-AUC, while class labels were obtained using the predefined classification threshold.

2.9.1 Linear and Kernel-Based Models

Logistic Regression, linear SVM, and radial basis function (RBF) SVM were evaluated. Logistic Regression used Elastic-Net regularisation. Linear SVM represented a linear decision boundary, while RBF SVM allowed nonlinear relationships.

The regularisation strength C was optimised for all three models. The Elastic-Net `l1_ratio` was also tuned for Logistic Regression, while the kernel parameter γ was tuned for RBF SVM. Balanced class weights were used.

2.9.2 Tree-Based Models

Five tree-based ensemble classifiers were evaluated: Random Forest, Extra Trees, Gradient Boosting, XGBoost, and LightGBM.

Random Forest and Extra Trees combine multiple independently fitted decision trees. Gradient Boosting, XGBoost, and LightGBM construct trees sequentially to correct errors made by earlier trees. Their hyperparameters controlled model size, tree depth, learning rate, feature sampling, observation sampling, and regularisation.

2.9.3 Distance-Based and Probabilistic Models

K-Nearest Neighbours (KNN) classified each ROI using nearby training observations in the standardised feature space. The number of neighbours, voting method, and distance metric were optimised.

Gaussian Naive Bayes estimated class probabilities from the distribution of each selected feature. Its variance-smoothing parameter was optimised to improve numerical stability.

All classifiers were evaluated using the same outer patient folds, selected features, SMOTE condition, and performance metrics. Classifier-specific hyperparameters were tuned independently within the inner cross-validation loop. Outer-test predictions were generated only after model selection had been completed.

2.10 Hyperparameter Optimisation Using Optuna

Classifier hyperparameters were optimised using Optuna within the inner loop of patient-aware nested cross-validation. This ensured that hyperparameter selection used only the current outer-training patients and did not use the outer-test data.

A separate Optuna study was created for each dataset, feature-selection strategy, SMOTE condition, classifier, and outer fold. Each study evaluated

$$N_{\text{trials}} = 10 \quad (28)$$

candidate hyperparameter settings. Candidates were generated using the Tree-structured Parzen Estimator sampler:

```
optuna.samplers.TPESampler(seed=100)
```

The objective was maximised, and a fixed random seed of 100 was used for reproducibility.

2.10.1 Inner-Loop Evaluation

Each trial was evaluated using up to three patient-aware inner folds from the current outer-training data. The actual number of folds was

$$K_{\text{inner,actual}} = \min(3, n_{\text{minority}}, n_{\text{patients}}), \quad (29)$$

with at least two valid folds required. All ROIs from one patient remained in the same inner partition.

Within each inner split, missing values were replaced using inner-training medians, low-variance features were removed, and standardisation was fitted using the inner-training data. When enabled, SMOTE was applied only to the transformed inner-training samples. The candidate classifier was then evaluated on the unchanged inner-validation data using ROC-AUC.

Predicted probabilities were used when available; otherwise, continuous decision scores were used. Inner folds containing only one class were excluded. Model-fitting or prediction failures were assigned an ROC-AUC of 0.5.

2.10.2 Optimisation Objective

For trial t , the inner-fold ROC-AUC values were combined using their median:

$$\tilde{A}_t = \text{median}_r(A_{t,r}), \quad (30)$$

where $A_{t,r}$ is the validation ROC-AUC for inner fold r .

The Optuna objective was

$$J_t = \tilde{A}_t - 0.001p, \quad (31)$$

where p is the number of selected features. Because the feature set was fixed within each Optuna study, this term was constant across its trials and did not change their ranking.

Chance-level performance was assigned when fewer than two valid folds were available, no valid ROC-AUC values remained, all features were removed, or model fitting failed.

2.10.3 Hyperparameter Search Spaces

The classifier-specific search spaces are shown in Table 2. Logarithmic sampling was used for parameters covering several orders of magnitude.

2.10.4 Final Outer-Fold Fitting

After the ten trials, the hyperparameter setting with the highest inner-loop objective was selected:

$$\lambda_m^{*(q)} = \arg \max_{\lambda \in \Lambda_m} J_m^{(q)}(\lambda), \quad (32)$$

where m identifies the classifier, q identifies the outer fold, and Λ_m is the corresponding search space.

A new classifier was then fitted to the complete processed outer-training data using the selected hyperparameters. When enabled, SMOTE was applied to the outer-training data before fitting. The model was evaluated once on the outer-test partition, which did not influence hyperparameter selection.

The selected hyperparameters, best inner objective value, feature count, and outer-test performance were saved for each classifier and outer fold.

Table 2: Hyperparameter search spaces used during Optuna optimisation. Log indicates logarithmic sampling.

Classifier	Hyperparameter	Search space
Logistic Regression	C l1_ratio	10^{-4} to 1, log 0 to 1
Linear SVM	C	10^{-4} to 1, log
RBF SVM	C γ	10^{-3} to 10, log 10^{-4} to 10^{-1} , log
Random Forest	n_estimators max_depth min_samples_leaf max_features	50, 100, 150, or 200 Integer from 2 to 8 Integer from 1 to 10 sqrt, log2, or all features
Extra Trees	n_estimators max_depth min_samples_leaf max_features	50 to 500 in increments of 50 Integer from 2 to 10 Integer from 2 to 20 sqrt, log2, or all features
Gradient Boosting	n_estimators max_depth learning_rate subsample	25, 50, 75, or 100 Integer from 1 to 3 0.01 to 0.10, log 0.70 to 1.00
K-Nearest Neighbours	n_neighbors weights metric	Integer from 3 to 15 Uniform or distance-based Euclidean or Manhattan
Gaussian Naive Bayes	var_smoothing	10^{-12} to 10^{-6} , log
XGBoost	n_estimators max_depth learning_rate subsample colsample_bytree reg_alpha reg_lambda	50 to 300 in increments of 50 Integer from 2 to 8 0.01 to 0.30, log 0.60 to 1.00 0.50 to 1.00 10^{-8} to 10, log 10^{-8} to 10, log
LightGBM	n_estimators max_depth num_leaves learning_rate subsample colsample_bytree reg_alpha reg_lambda	50 to 300 in increments of 50 Integer from 2 to 8 Integer from 8 to 64 0.01 to 0.30, log 0.60 to 1.00 0.50 to 1.00 10^{-8} to 10, log 10^{-8} to 10, log

2.11 Model Performance Evaluation and Uncertainty

Model performance was evaluated using predictions from the outer-test folds of the patient-aware nested cross-validation procedure. Each ROI received one out-of-fold prediction from a model that had not been trained using that ROI or any other ROI from the same patient.

The primary evaluation measure was the mean outer-fold ROC-AUC. Additional discrimination, classification, agreement, and calibration measures were also calculated.

2.11.1 Primary and Pooled Performance Measures

ROC-AUC measures how well the classifier ranks malignant ROIs above benign ROIs across different thresholds. ROC-AUC was calculated separately for each of the five outer folds, and the primary score was

$$\overline{\text{AUC}} = \frac{1}{K_{\text{outer}}} \sum_{q=1}^{K_{\text{outer}}} \text{AUC}^{(q)}, \quad K_{\text{outer}} = 5. \quad (33)$$

The standard deviation across folds was also reported to describe variation between different test-patient groups.

Predictions from all five outer folds were additionally combined to calculate pooled ROC-AUC:

$$\text{AUC}_{\text{pooled}} = \text{AUC}(\mathbf{y}, \hat{\mathbf{p}}_{\text{OOF}}). \quad (34)$$

Mean outer-fold and pooled ROC-AUC were reported separately because they summarise performance differently. Precision–recall performance was evaluated using average precision, reported as PR-AUC, both fold-wise and using pooled out-of-fold predictions.

2.11.2 Threshold-Dependent Metrics

For probability-based predictions, an ROI was classified as malignant when

$$\hat{p}_i \geq 0.5, \quad (35)$$

and as benign otherwise. The threshold was fixed before evaluation.

The resulting predictions were summarised using accuracy, sensitivity, specificity, precision, balanced accuracy, F1-score, and Cohen’s kappa. Balanced accuracy was calculated as

$$\text{Balanced Accuracy} = \frac{\text{Sensitivity} + \text{Specificity}}{2}, \quad (36)$$

and the F1-score as

$$F_1 = 2 \frac{\text{Precision} \times \text{Sensitivity}}{\text{Precision} + \text{Sensitivity}}. \quad (37)$$

These metrics were calculated separately for each outer fold and again from the pooled out-of-fold predictions.

2.11.3 Probability Calibration

Calibration was evaluated using the pooled out-of-fold probabilities. Predicted probabilities were grouped and compared with the observed proportion of malignant ROIs.

Calibration error was summarised using the Brier score:

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2. \quad (38)$$

Lower Brier scores indicate better agreement between predicted probabilities and observed outcomes.

2.11.4 Patient-Clustered Confidence Intervals

Uncertainty in pooled performance measures was estimated using patient-cluster bootstrap resampling. Patients, rather than individual ROIs, were sampled with replacement so that all ROIs from the same patient remained together.

A total of

$$B = 2000 \quad (39)$$

bootstrap samples were generated. Samples containing only one class were excluded when the corresponding metric could not be calculated.

The 95% confidence interval was obtained using the percentile method:

$$CI_{95\%} = [Q_{0.025}, Q_{0.975}]. \quad (40)$$

Confidence intervals were calculated for pooled ROC-AUC, PR-AUC, accuracy, balanced accuracy, precision, sensitivity, specificity, F1-score, Cohen’s kappa, and Brier score.

2.11.5 Model Ranking

Candidate pipelines were ranked primarily by mean outer-fold ROC-AUC. The fold-level standard deviation, pooled ROC-AUC, PR-AUC, balanced accuracy, and other metrics were used as complementary measures.

The highest-ranked overall model was examined in greater detail using pooled ROC and precision–recall curves, a confusion matrix, calibration analysis, and fold-level performance summaries.

2.12 Feature-Selection Stability and Model Interpretation

Feature-selection stability was examined to determine how often the same predictors were selected across different outer cross-validation folds. Model interpretation was then performed using Shapley Additive Explanations (SHAP).

These two analyses were treated separately. Selection stability measured how consistently a feature was retained, while SHAP measured how strongly a feature influenced the output of the fitted interpretation model.

2.12.1 Feature-Selection Stability

For feature j , selection frequency was calculated as

$$f_j = \frac{1}{K_{\text{outer}}} \sum_{q=1}^{K_{\text{outer}}} \mathbb{I}(j \in \mathcal{F}^{(q)}), \quad (41)$$

where $K_{\text{outer}} = 5$ and $\mathcal{F}^{(q)}$ is the feature set selected in outer fold q .

A frequency of 1 indicated selection in all five folds, while values of 0.8, 0.6, 0.4, and 0.2 indicated selection in four, three, two, and one fold, respectively.

Features were ranked by selection frequency. For multi-source datasets, the source prefix was retained so that features from ADC, Pre, Post1, Post2, and T2 could be tracked separately.

2.12.2 Recurrent Features for Interpretation

SHAP analysis focused on features selected in at least two outer folds:

$$c_j \geq \max(2, \lceil 0.40K_{\text{outer}} \rceil), \quad (42)$$

where c_j is the number of folds in which feature j was selected. With five outer folds, this corresponded to a minimum selection frequency of 40%.

If no feature met this criterion, the most frequently selected features were used instead.

2.12.3 Interpretation Model

For the selected experimental configuration, the classifier with the highest mean outer-fold ROC-AUC was used for interpretation.

A separate model was refitted using the complete dataset and only the recurrent features. Missing values were replaced using full-cohort medians, and the features were standardised.

Classifier hyperparameters selected across the outer folds were combined without using outer-test performance. Numerical parameters were summarised using their median, while categorical parameters were summarised using their most frequent value.

This full-cohort model was used only for interpretation. All predictive performance measures remained based on the patient-aware out-of-fold predictions.

2.12.4 SHAP Analysis

SHAP was used to explain how the recurrent features influenced the malignant-class model output. For ROI i ,

Table 3: Performance measures used for model evaluation. Threshold-dependent measures used a classification threshold of 0.5.

Measure	Preferred value	Interpretation
ROC-AUC	Higher	Ability to rank malignant ROIs above benign ROIs.
PR-AUC	Higher	Relationship between sensitivity and precision across thresholds.
Accuracy	Higher	Proportion of correctly classified ROIs.
Balanced accuracy	Higher	Average of sensitivity and specificity.
Sensitivity	Higher	Proportion of malignant ROIs correctly identified.
Specificity	Higher	Proportion of benign ROIs correctly identified.
Precision	Higher	Proportion of predicted malignant ROIs that were malignant.
F1-score	Higher	Combined measure of precision and sensitivity.
Cohen’s kappa	Higher	Agreement between predicted and observed classes beyond chance.
Brier score	Lower	Error between predicted probabilities and observed outcomes.

$$h(\mathbf{x}_i) = \phi_0 + \sum_{j=1}^{p_{\text{XAI}}} \phi_{ij}, \quad (43)$$

where ϕ_0 is the baseline model output and ϕ_{ij} is the SHAP contribution of feature j .

Positive SHAP values moved the model output towards the malignant class, while negative values moved it towards the benign class.

Global feature importance was calculated using the mean absolute SHAP value:

$$I_j^{\text{SHAP}} = \frac{1}{n} \sum_{i=1}^n |\phi_{ij}|. \quad (44)$$

Features with larger values had a greater average influence on the model output.

2.12.5 Interpretation Outputs

Two SHAP plots were generated. A bar plot ranked features by mean absolute SHAP value, while a summary dot plot showed both the magnitude and direction of feature effects across ROIs.

The selected-feature frequencies, SHAP importance values, plots, fitted interpretation model, preprocessing objects, and aggregated hyperparameters were saved for reproducibility.

SHAP values were interpreted only as explanations of the fitted model. They do not demonstrate causal or biological relationships.

2.13 Software Implementation and Reproducibility

The complete workflow was implemented in Python using Jupyter notebooks. The final notebook environment used Python 3.13.9.

NumPy, pandas, and SciPy were used for numerical and data-processing operations. Scikit-learn was used for preprocessing, cross-validation, machine-learning models, and performance metrics. SMOTE was implemented using imbalanced-learn, and Optuna was used for hyperparameter optimisation. XGBoost and LightGBM provided the corresponding boosted-tree classifiers. SHAP was used for model interpretation, Matplotlib for figures, and Joblib for saving fitted models and preprocessing objects.

A fixed random seed of 100 was used whenever supported, including cross-validation splitting, feature selection, SMOTE, Optuna, stochastic classifiers, and bootstrap calculations. This improved the reproducibility of repeated runs.

2.13.1 Automated Notebook Execution

The analysis was organised into parameterised modelling notebooks and separate batch-runner notebooks. Parameters such as the input dataset, dataset name, output directory, and SMOTE condition were supplied to the analysis notebooks au-

tomatically.

Papermill was used to run the same modelling code across different dataset configurations. Separate batch runners were used for single-source, pairwise, triple-source, quadruple-source, and five-source experiments.

Before execution, the runners checked the required input files and notebook parameters. Invalid or missing datasets were not processed.

2.13.2 Batch Execution

The workflow supported both interactive and batch execution. Publication and diagnostic plots could be produced during interactive analysis but were skipped during large batch runs. This affected only figure generation and did not change preprocessing, feature selection, cross-validation, model optimisation, or performance calculation.

To reduce problems caused by nested parallel processing, batch runs were limited to a single computational worker where required. This provided a more stable and consistent environment for running the full experimental grid.

2.13.3 Output Management and Traceability

Each experimental run used a separate output directory. The workflow saved the main information required to reproduce and review the analysis, including patient-fold assignments, selected features, hyperparameters, fold-level and pooled predictions, performance metrics, confidence intervals, feature-stability results, and summary tables.

Tabular results were mainly stored as CSV files, while configuration information was stored in JSON format. Fitted models and preprocessing objects were saved using Joblib. Executed notebooks and batch logs were also retained to record the parameters and execution status of each run.

After each experimental group was completed, the individual outputs were combined into standardised summary tables for comparison across datasets, feature-selection strategies, SMOTE conditions, and classifiers.

The same core workflow was applied to all 31 dataset configurations. Only the input datasets and source combinations changed, while the patient-aware validation design, feature-selection strategies, candidate feature counts, classifier set, Op-tuna trial budget, random seed, and primary evaluation metric remained fixed.

3 Results

3.1 Dataset Characteristics and Quality-Control Results

All 31 planned dataset configurations were successfully included in the analysis: five single-source, ten pairwise, ten triple-source, five quadruple-source, and one five-source configuration.

The five single-source datasets represented 48 patients and contained 114–117 ROIs with 144–145 numerical predictor features. The numbers of benign and malignant ROIs were similar across sources.

Combining multiple sources increased the number of predictor features but did not increase the number of patients. Because only ROIs shared by all sources in a combination were retained, the sample size varied slightly between configurations. Dataset dimensions at each fusion level are summarised in Table ??.

The complete five-source dataset contained 113 matched ROIs and 724 predictor features.

A detailed quality-control review of the ADC dataset identified 114 ROIs from 48 patients. Patients contributed between 1 and 7 ROIs. Twelve missing values were found across six feature columns, while no infinite values, constant features, duplicated rows, duplicated ROI identifiers, or duplicated patient–ROI pairs were detected.

Potential outliers were retained, and partially missing observations were not removed. Missing values were later handled using training-fold median imputation. Patients with multiple ROIs, including those with ROIs from both classes, were retained while all ROIs from the same patient remained together during cross-validation.

3.2 Overall Performance Across the 31 Dataset Configurations

All 31 source configurations were evaluated using the same patient-aware nested cross-validation framework. For each configuration, the highest-ranked pipeline was selected according to mean outer-fold ROC-AUC.

Figure 2 compares the best pipeline for each source configuration. Error bars show the standard deviation of ROC-AUC across the five outer folds.

The best overall result was obtained with the ADC–Post1–Pre triple-source configuration. Its pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and linear SVM, and achieved

$$\text{Mean ROC-AUC} = 0.836 \pm 0.088, \quad (45)$$

with a pooled ROC-AUC of 0.799. Auto- k selected an average of 9.6 features across the five folds.

The ADC–Post2–T2 triple combination produced a very similar mean ROC-AUC of 0.834 ± 0.045 , with a lower fold-to-fold variation and a pooled ROC-AUC of 0.756.

The highest-ranked single-source, pairwise, and quadruple-source configurations were Post2, ADC–T2, and ADC–Post2–Pre–T2, respectively. Their mean ROC-AUC values were 0.790, 0.799, and 0.802. The complete five-source configuration achieved a lower mean ROC-AUC of 0.747.

These results show that adding more sources did not continuously improve classification performance. The highest performance was obtained using a selected three-source combination rather than the complete five-source dataset.

Table 4 summarises the best pipeline at each fusion level.

3.3 Single-Source Classification Results

The five single-source datasets achieved mean outer-fold ROC-AUC values from 0.674 to 0.790 (Figure 2a).

Post2 produced the best single-source result. Its highest-ranked pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and an RBF SVM. It achieved a mean ROC-AUC of 0.790 ± 0.127 and a pooled ROC-AUC of 0.744. The corresponding mean and pooled PR-AUC values were 0.876 and 0.815.

Post1 ranked second with a mean ROC-AUC of 0.767 ± 0.098 and produced the highest pooled ROC-AUC among the single-source datasets (0.765). ADC, Pre, and T2 achieved mean ROC-AUC values of 0.764, 0.698, and 0.674, respectively.

Four of the five highest-ranked single-source pipelines used SMOTE, while Post2 was the only winner without SMOTE. The complete results are summarised in Table 5.

3.4 Pairwise Classification Results

The ten pairwise combinations produced mean outer-fold ROC-AUC values from 0.665 to 0.799 (Figure 2b).

ADC–T2 achieved the highest mean ROC-AUC, 0.799 ± 0.084 , using Elastic Net followed by mRMR, no SMOTE, and an RBF SVM. Its pooled ROC-AUC was 0.722.

ADC–Pre ranked second with 0.792 ± 0.035 and showed the lowest fold-to-fold variation among the pairwise winners. ADC–Post1 ranked third with 0.788 ± 0.050 and produced the highest pooled ROC-AUC of 0.775.

The four highest-ranked pairwise combinations all included ADC. Elastic Net followed by mRMR was selected in seven of the ten configurations, and seven winners used SMOTE. RBF SVM was the most frequent winning classifier. Full results are shown in Table 6.

3.5 Triple-Source Classification Results

The ten triple-source configurations achieved mean outer-fold ROC-AUC values from 0.682 to 0.836 (Figure 2c).

ADC–Post1–Pre achieved the highest performance among the triple-source datasets and across

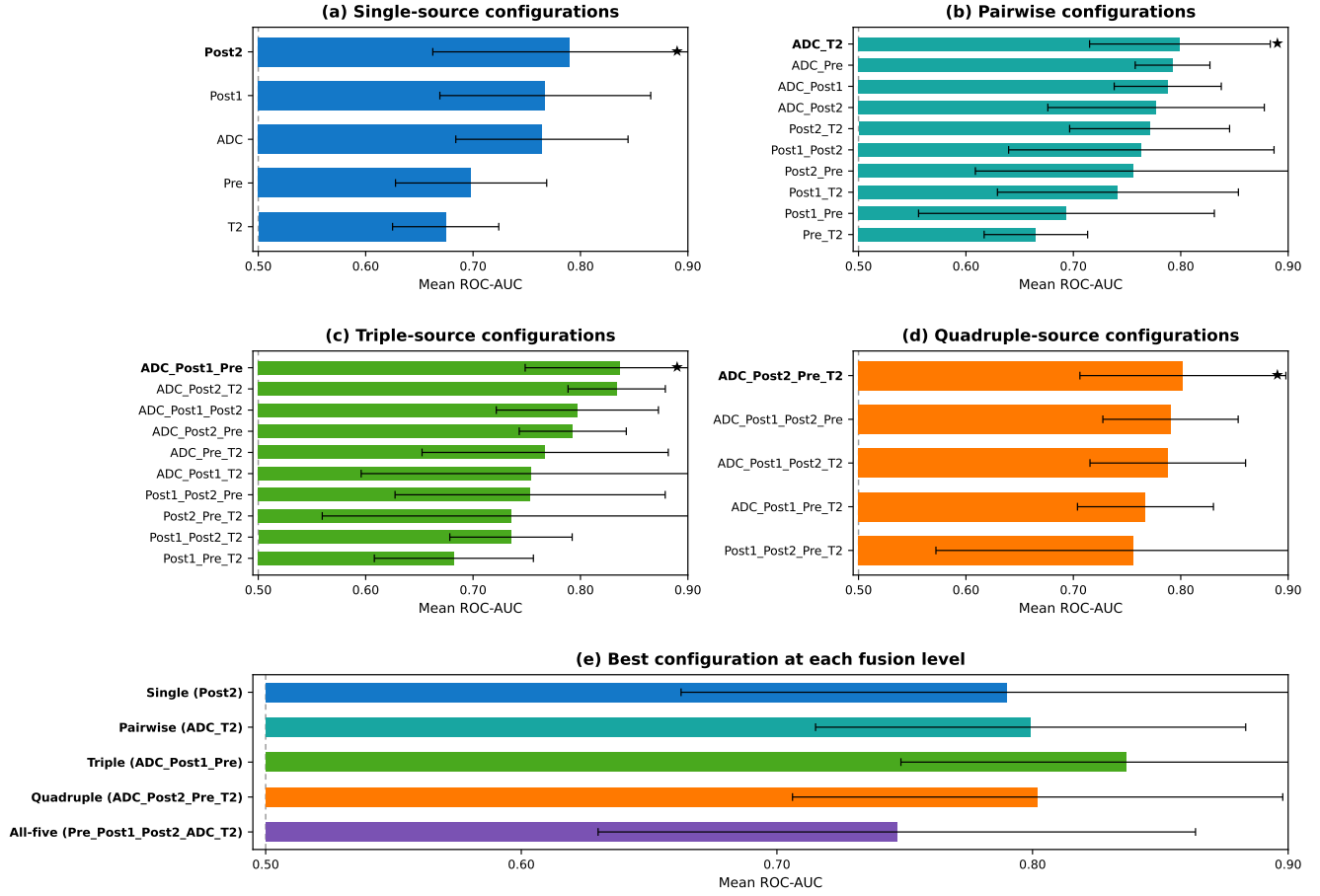


Figure 2: **Classification performance across the 31 radiomic source configurations.** The best pipeline for each configuration is shown according to mean ROC-AUC across five patient-aware outer folds. Error bars represent the corresponding standard deviation. (a) Single-source, (b) pairwise, (c) triple-source, (d) quadruple-source configurations, and (e) the best configuration at each fusion level. The dashed line indicates chance-level ROC-AUC of 0.50.

Table 4: Highest-ranked configuration at each feature-fusion level. Configurations were selected according to mean ROC-AUC across the five patient-aware outer folds. Pooled ROC-AUC, PR-AUC, and balanced accuracy were calculated from the combined out-of-fold predictions. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; Bal. Acc., balanced accuracy; SD, standard deviation.

Fusion level	Dataset/configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
Single	Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 $\pm$ 0.127	0.744	0.815	0.669	10.8
Pairwise	ADC-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.799 $\pm$ 0.084	0.722	0.769	0.637	10.4
Triple	ADC-Post1-Pre	EN $\rightarrow$ mRMR	No	Linear SVM	0.836 $\pm$ 0.088	0.799	0.832	0.713	9.6
Quadruple	ADC-Post2-Pre-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.802 $\pm$ 0.096	0.723	0.797	0.700	13.2
All five	ADC-Post1-Post2-Pre-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.747 $\pm$ 0.117	0.724	0.792	0.595	12.4

Table 5: Highest-ranked pipeline for each single-source dataset. Pipelines were ranked by mean ROC-AUC across five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; KNN, K-Nearest Neighbours; Bal. Acc., balanced accuracy; SD, standard deviation.

Dataset	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
Post2	EN $\rightarrow$ mRMR	No	RBF SVM	0.790 $\pm$ 0.127	0.744	0.815	0.669	10.8
Post1	EN $\rightarrow$ mRMR	Yes	KNN	0.767 $\pm$ 0.098	0.765	0.815	0.705	7.2
ADC	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.764 $\pm$ 0.080	0.691	0.720	0.683	8.8
Pre	mRMR $\rightarrow$ EN	Yes	Logistic Regression	0.698 $\pm$ 0.070	0.679	0.740	0.594	10.4
T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.674 $\pm$ 0.049	0.642	0.679	0.590	12.4

Table 6: Highest-ranked pipeline for each pairwise source configuration. Pipelines were ranked by mean ROC-AUC across five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; KNN, K-Nearest Neighbours; NB, Gaussian Naive Bayes; Bal. Acc., balanced accuracy; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
ADC-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.799 $\pm$ 0.084	0.722	0.769	0.637	10.4
ADC-Pre	EN $\rightarrow$ mRMR	No	Logistic Regression	0.792 $\pm$ 0.035	0.723	0.767	0.674	9.6
ADC-Post1	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.788 $\pm$ 0.050	0.775	0.821	0.709	8.0
ADC-Post2	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.777 $\pm$ 0.101	0.687	0.783	0.671	8.4
Post2-T2	mRMR $\rightarrow$ EN	Yes	XGBoost	0.771 $\pm$ 0.074	0.743	0.786	0.671	8.4
Post1-Post2	mRMR $\rightarrow$ EN	Yes	RBF SVM	0.763 $\pm$ 0.124	0.744	0.782	0.680	5.6
Post2-Pre	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.756 $\pm$ 0.147	0.748	0.826	0.687	10.4
Post1-T2	EN $\rightarrow$ mRMR	Yes	KNN	0.742 $\pm$ 0.112	0.718	0.773	0.651	11.2
Post1-Pre	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.694 $\pm$ 0.138	0.687	0.740	0.648	10.4
Pre-T2	EN $\rightarrow$ mRMR	No	Gaussian NB	0.665 $\pm$ 0.048	0.633	0.676	0.552	9.6

all 31 source configurations. Its pipeline used Elastic Net followed by mRMR, no SMOTE, and a linear SVM. It achieved a mean ROC-AUC of 0.836 ± 0.088 and a pooled ROC-AUC of 0.799. Mean and pooled PR-AUC were 0.879 and 0.832, respectively, and pooled balanced accuracy was 0.713.

ADC-Post2-T2 ranked second with 0.834 ± 0.045 . Its lower standard deviation indicated more consistent performance across the five outer folds, although its pooled ROC-AUC was lower at 0.756.

ADC-Post1-Post2 ranked third with 0.797 ± 0.075 and a pooled ROC-AUC of 0.790.

The six highest-ranked triple-source combinations all included ADC. Elastic Net followed by mRMR was selected in six of the ten configurations. Five winners used SMOTE and five did not. The complete results are reported in Table 7.

3.6 Quadruple-Source Classification Results

The five quadruple-source configurations produced mean outer-fold ROC-AUC values from 0.756 to 0.802 (Figure 2d).

ADC-Pre-Post2-T2 achieved the highest mean ROC-AUC, 0.802 ± 0.096 , using Elastic Net followed by mRMR, SMOTE, and an RBF SVM. Its pooled ROC-AUC was 0.723.

ADC-Pre-Post1-Post2 ranked second with 0.790 ± 0.063 , while ADC-Post1-Post2-T2 ranked third with 0.788 ± 0.072 and produced the highest pooled ROC-AUC among the quadruple-source datasets (0.770).

All five winning pipelines used Elastic Net followed by mRMR, and four used SMOTE. The four combinations containing ADC ranked above the configuration without ADC. Results are summarised in Table 8.

3.7 Complete Five-Source Classification Result

The complete five-source dataset contained 113 matched ROIs and 724 predictor features. Its highest-ranked pipeline used Elastic Net followed

by mRMR, Auto- k , no SMOTE, and an RBF SVM.

The model achieved a mean outer-fold ROC-AUC of 0.747 ± 0.117 and a pooled ROC-AUC of 0.724 (95% patient-clustered CI: 0.602–0.825). Mean and pooled PR-AUC were 0.810 and 0.792, respectively.

The pooled balanced accuracy was 0.595. At the fixed threshold of 0.5, pooled sensitivity was 0.810, precision was 0.622, and the F1-score was 0.703. The pooled Cohen’s kappa and Brier score were 0.197 and 0.220, respectively.

Auto- k selected an average of 12.4 features, and 38 different features were selected in at least one outer fold.

As shown in Figure 2(e), the five-source model performed below the best triple-source model. Its mean ROC-AUC was 0.089 lower than that of ADC-Post1-Pre, showing that combining all available feature sources did not provide the best classification performance.

3.8 Comparison of Feature-Selection Strategies

The three feature-selection strategies were compared across all 31 source configurations. For each configuration and strategy, the best result was retained after comparing the SMOTE and no-SMOTE conditions and all ten classifiers.

Figure 3 summarises the distribution of mean outer-fold ROC-AUC values and the number of configuration-level wins for each strategy.

Elastic Net followed by mRMR produced the strongest overall results. Its mean and median ROC-AUC values were 0.754 and 0.760, compared with 0.736 and 0.747 for mRMR followed by Elastic Net, and 0.694 and 0.698 for correlation filtering followed by mRMR.

Elastic Net followed by mRMR ranked first in 20 of the 31 source configurations, while mRMR followed by Elastic Net ranked first in 11. Correlation filtering followed by mRMR did not produce the highest-ranked result for any configuration.

The advantage of Elastic Net followed by mRMR was also observed across most fusion lev-

Table 7: Highest-ranked pipeline for each triple-source configuration. Pipelines were ranked by mean ROC-AUC across five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; NB, Gaussian Naive Bayes; Bal. Acc., balanced accuracy; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
ADC-Post1-Pre	EN $\rightarrow$ mRMR	No	Linear SVM	0.836 $\pm$ 0.088	0.799	0.832	0.713	9.6
ADC-Post2-T2	EN $\rightarrow$ mRMR	No	Linear SVM	0.834 $\pm$ 0.045	0.756	0.813	0.718	10.4
ADC-Post1-Post2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.797 $\pm$ 0.075	0.790	0.841	0.719	7.6
ADC-Post2-Pre	mRMR $\rightarrow$ EN	Yes	Logistic Regression	0.793 $\pm$ 0.050	0.722	0.777	0.683	14.4
ADC-Pre-T2	EN $\rightarrow$ mRMR	Yes	LightGBM	0.767 $\pm$ 0.115	0.720	0.778	0.648	13.2
ADC-Post1-T2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.754 $\pm$ 0.158	0.765	0.836	0.695	5.6
Post1-Post2-Pre	mRMR $\rightarrow$ EN	Yes	Linear SVM	0.753 $\pm$ 0.126	0.715	0.793	0.686	11.6
Post2-Pre-T2	EN $\rightarrow$ mRMR	No	RBF SVM	0.735 $\pm$ 0.176	0.707	0.783	0.637	13.2
Post1-Post2-T2	mRMR $\rightarrow$ EN	No	Gaussian NB	0.735 $\pm$ 0.057	0.706	0.766	0.613	10.0
Post1-Pre-T2	mRMR $\rightarrow$ EN	No	Gaussian NB	0.682 $\pm$ 0.074	0.695	0.744	0.623	11.6

Table 8: Highest-ranked pipeline for each quadruple-source configuration. Pipelines were ranked by mean ROC-AUC across five patient-aware outer folds. EN, Elastic Net; mRMR, minimum redundancy maximum relevance; SVM, support vector machine; Bal. Acc., balanced accuracy; SD, standard deviation.

Configuration	Feature selection	SMOTE	Classifier	Mean AUC $\pm$ SD	Pooled AUC	Pooled PR-AUC	Pooled Bal. Acc.	Mean k
ADC-Post2-Pre-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.802 $\pm$ 0.096	0.723	0.797	0.700	13.2
ADC-Post1-Post2-Pre	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.790 $\pm$ 0.063	0.765	0.795	0.719	6.0
ADC-Post1-Post2-T2	EN $\rightarrow$ mRMR	Yes	Logistic Regression	0.788 $\pm$ 0.072	0.770	0.822	0.691	8.0
ADC-Post1-Pre-T2	EN $\rightarrow$ mRMR	No	Linear SVM	0.767 $\pm$ 0.063	0.725	0.795	0.667	9.6
Post1-Post2-Pre-T2	EN $\rightarrow$ mRMR	Yes	RBF SVM	0.756 $\pm$ 0.184	0.713	0.745	0.665	15.2

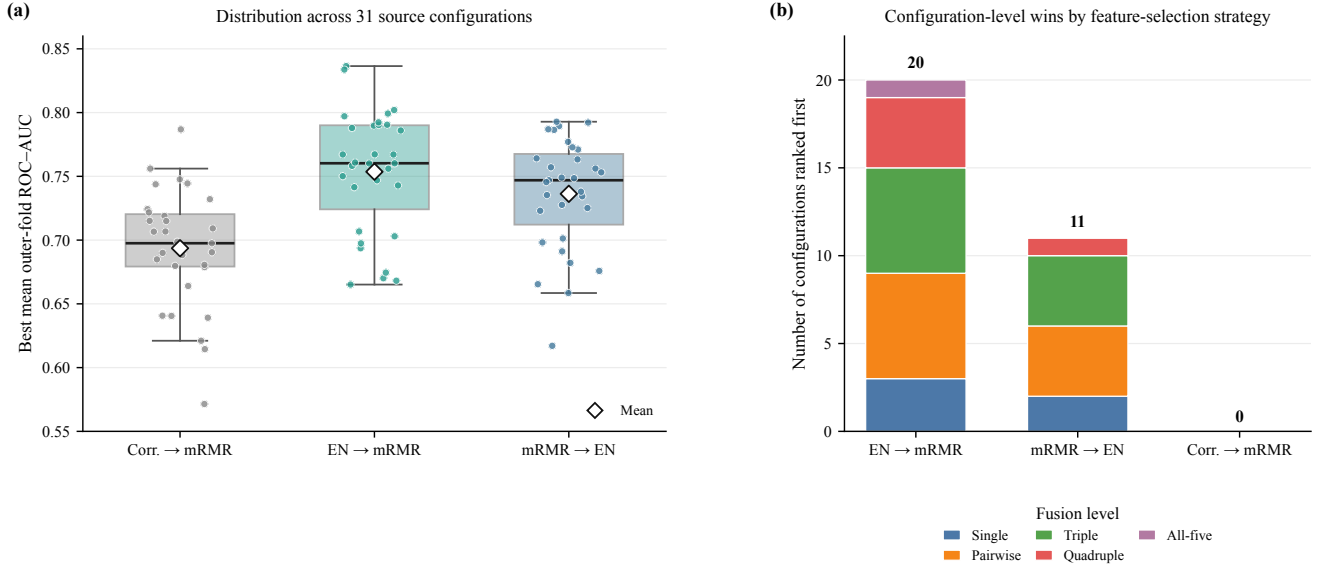


Figure 3: Comparison of feature-selection strategies across the 31 source configurations. (a) Distribution of the best mean outer-fold ROC-AUC obtained for each strategy. Points represent source configurations, boxes show the interquartile range, horizontal lines indicate medians, and white diamonds indicate means. (b) Number of configurations in which each strategy achieved the highest mean ROC-AUC, separated by feature-fusion level. EN, Elastic Net; mRMR, minimum redundancy maximum relevance.

els. It ranked first for three of five single-source, six of ten pairwise, six of ten triple-source, four of five quadruple-source, and the complete five-source configuration.

The highest ROC-AUC obtained with Elastic Net followed by mRMR was 0.836 for ADC–Post1–Pre. The corresponding maximum values for mRMR followed by Elastic Net and correlation filtering followed by mRMR were 0.793 and 0.787, respectively.

3.9 Effect of SMOTE on Classification Performance

The effect of SMOTE was evaluated across all 31 source configurations. For each configuration, the best SMOTE and best no-SMOTE pipelines were identified after comparison of the three feature-selection strategies and ten classifiers.

The difference between the two conditions was calculated as

$$\Delta AUC_d = AUC_{d,SMOTE} - AUC_{d,no-SMOTE}. \quad (46)$$

Positive values indicate better performance with SMOTE, while negative values indicate better performance without SMOTE.

SMOTE produced the higher mean outer-fold ROC-AUC in 19 of the 31 configurations, while no-SMOTE performed better in 12. However, the overall difference was small. The average of the best SMOTE results was 0.760 compared with 0.757 without SMOTE, giving an average difference of only +0.003.

For 26 of the 31 configurations, the absolute difference between the two conditions was below 0.01 ROC-AUC. The largest improvement with SMOTE occurred for Post1–Post2–Pre, increasing from 0.728 to 0.753 (+0.025). The largest decrease occurred for ADC–Post1–Pre, where performance changed from 0.836 without SMOTE to 0.823 with SMOTE (−0.013).

SMOTE performed better in four of five single-source configurations, five of ten pairwise configurations, five of ten triple-source configurations, and all five quadruple-source configurations. The no-SMOTE condition performed better for the

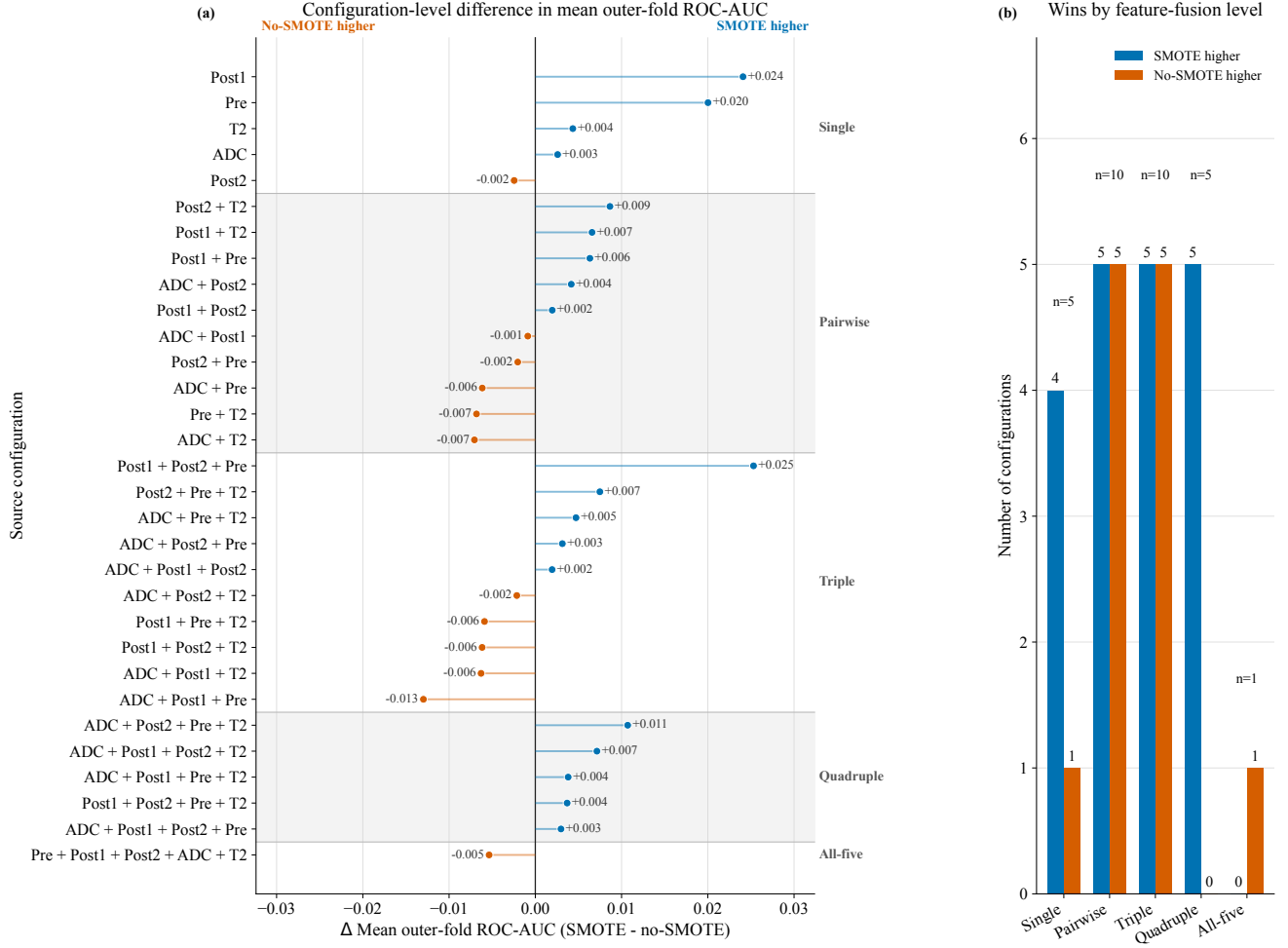


Figure 4: **Effect of SMOTE across the 31 source configurations.** (a) Difference in mean outer-fold ROC-AUC between the best SMOTE and no-SMOTE pipelines for each configuration. Positive values favour SMOTE and negative values favour no-SMOTE. (b) Number of configuration-level wins for each condition at every fusion level.

complete five-source dataset.

Overall, the effect of SMOTE was small and depended on the source configuration. It was therefore kept as an optional experimental condition rather than applied to all datasets.

3.10 Comparison of Machine-Learning Classifiers

Classifier selection was examined across the 31 highest-ranked configuration-level pipelines. For each source configuration, only the classifier belonging to the pipeline with the highest mean outer-fold ROC-AUC was counted.

RBF SVM was the most frequently selected classifier, appearing in 11 of the 31 highest-ranked pipelines (35.5%). Logistic Regression was selected seven times (22.6%), and linear SVM six times (19.4%). Together, the two SVM variants accounted for 17 of the 31 winning pipelines.

Gaussian Naive Bayes was selected in three configurations, K-Nearest Neighbours in two, and XGBoost and LightGBM once each. Random Forest, Extra Trees, and Gradient Boosting were not selected as the highest-ranked classifier for any source configuration.

Classifier preferences also varied with fusion level. RBF SVM was most common among pairwise and quadruple-source configurations, while Logistic Regression and linear SVM were more frequent among the triple-source configurations. RBF SVM was also selected for the complete five-source dataset.

These counts describe how often each classifier appeared in the best pipeline for a source configuration. They do not represent the average performance of each classifier across all experimental conditions.

3.11 Detailed Performance of the Highest-Ranked Model

The highest-ranked configuration was ADC–Post1–Pre. Its pipeline used Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM.

Figure 6 shows the pooled out-of-fold performance of this model. All predictions were generated for outer-test patients who were not used during the corresponding model-development stage.

The model achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a pooled ROC-AUC of 0.799. Mean and pooled PR-AUC were 0.879 and 0.832, respectively.

At the fixed classification threshold of 0.5, 81 of 113 ROIs were classified correctly, giving a pooled accuracy of 0.717. The confusion matrix contained 34 true negatives, 16 false positives, 16 false negatives, and 47 true positives.

The corresponding sensitivity and precision were both 0.746, specificity was 0.680, balanced accuracy was 0.713, and the F1-score was 0.746. Cohen’s kappa was 0.426, and the pooled Brier score was 0.183.

Auto- k selected an average of 9.6 features across the five outer folds, with feature counts ranging from 4 to 18. In total, 28 different features were selected in at least one fold. No feature appeared in all five folds, while the most frequently selected feature appeared in four.

Feature-selection stability and SHAP-based interpretation are reported in Section 3.12.

3.12 Feature-Selection Stability and Model Interpretation

Feature-selection stability and SHAP interpretation were examined for the highest-ranked ADC–Post1–Pre pipeline, which used Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM.

Feature selection was performed independently in each of the five outer folds. Auto- k selected an average of 9.6 features per fold, and 28 different features were selected at least once. Features selected in at least two folds were considered recurrent.

Thirteen features met this criterion (Figure 7(a)). No feature was selected in all five folds. The most stable feature was the ADC intensity-histogram median, which appeared in four folds. Five other features appeared in three

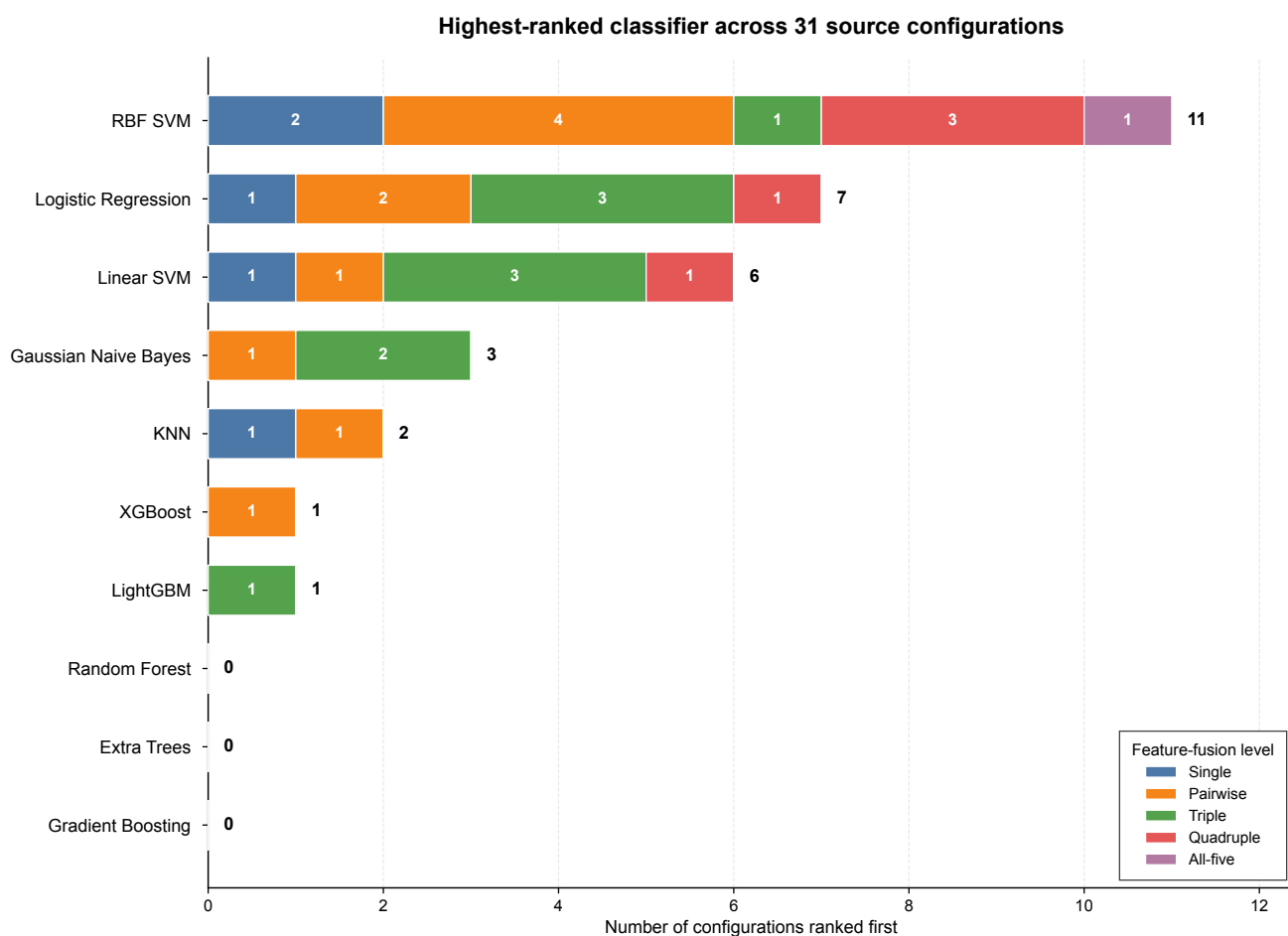


Figure 5: Classifier frequency among the highest-ranked pipelines across the 31 source configurations. Horizontal stacked bars show the number of configuration-level selections contributed by each feature-fusion level. Classifiers with no wins are retained to show all ten evaluated models. KNN, K-Nearest Neighbours; SVM, support vector machine.

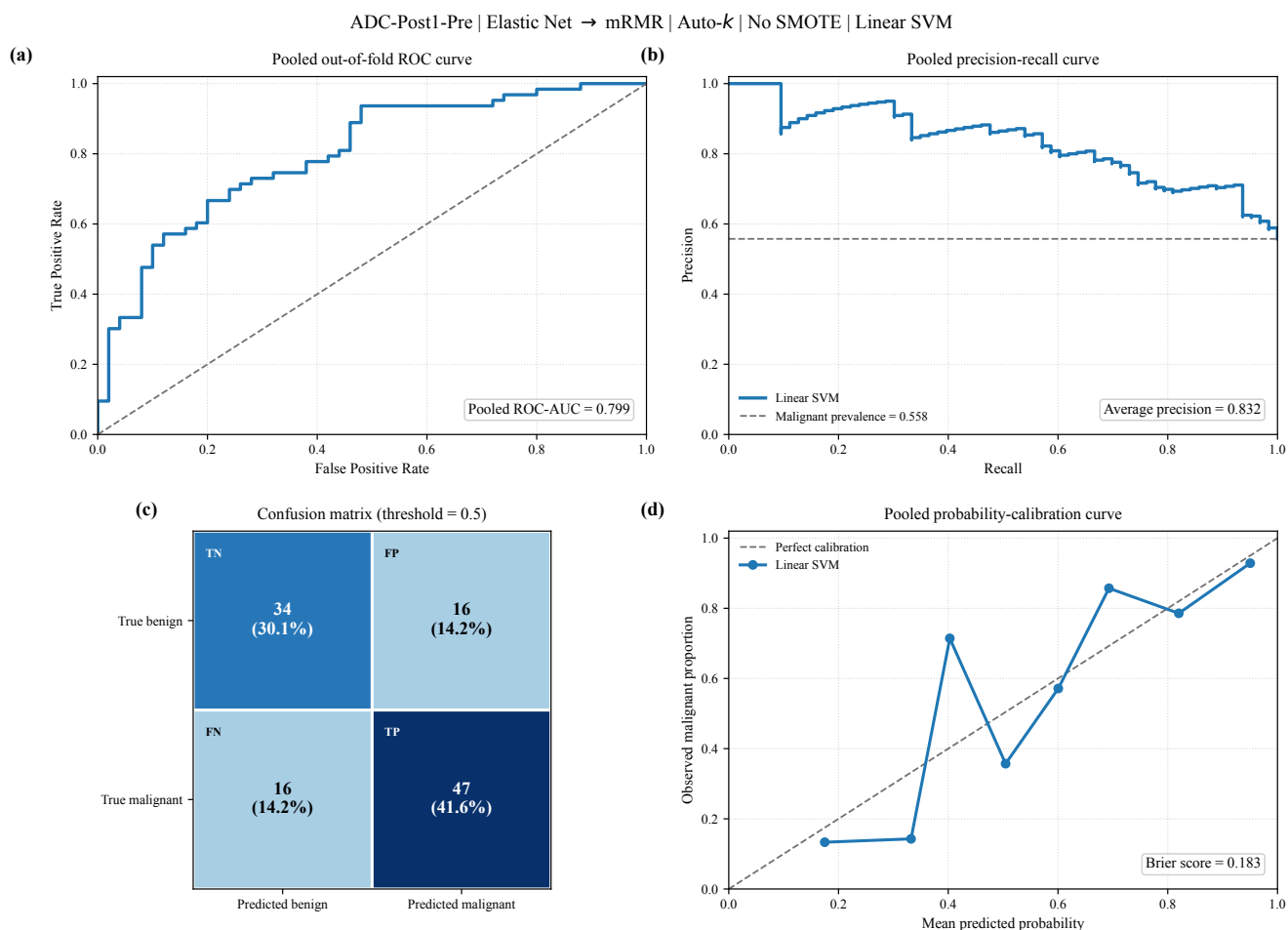


Figure 6: **Detailed out-of-fold performance of the highest-ranked pipeline.** The model used ADC, Post1, and Pre features with Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM. (a) Pooled ROC curve. (b) Pooled precision–recall curve. (c) Confusion matrix at the predefined threshold of 0.5. (d) Probability-calibration curve. All results were calculated from pooled predictions across the five patient-aware outer-test folds. AUC, area under the curve; mRMR, minimum redundancy maximum relevance; SVM, support vector machine.

fold, while the remaining recurrent features appeared in two. All recurrent features originated from ADC or Post1; no Pre-derived feature was selected in at least two folds.

For SHAP analysis, a separate linear SVM was refitted using the recurrent features and the complete cohort. This model was used only for interpretation and did not contribute to the reported performance estimates.

As shown in Figure 7(b), the ADC intensity-histogram median and ADC intensity-based median had the largest mean absolute SHAP values. Two Post1 intensity-histogram features followed them, while several Post1 morphological and texture features also contributed to the model.

The SHAP summary plot (Figure 7(c)) shows the direction of these effects. Higher values of the leading ADC intensity features and the main Post1 intensity-histogram features were generally associated with higher malignant-class model outputs. In contrast, higher values of some Post1 morphological features, including Compactness and Sphericity, were more often associated with lower malignant-class outputs.

Feature-selection frequency and SHAP importance describe different properties. Selection frequency measures how consistently a feature was retained across patient groups, whereas SHAP measures its influence within the fitted interpretation model. SHAP results were therefore used to explain model behaviour and not as evidence of causal or biological effects.

4 Discussion

4.1 Principal Findings

This study compared all 31 possible combinations of five MRI-derived radiomic feature sources using the same patient-aware machine-learning framework. The main finding was that combining all available sources did not give the best result. Instead, the highest performance was obtained from a selected triple-source combination.

The best pipeline used ADC, Post1, and Pre features with Elastic Net followed by mRMR, Auto- k ,

no SMOTE, and a linear SVM. It achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 . A second triple-source combination, ADC-Post2-T2, achieved a very similar result of 0.834 ± 0.045 . In comparison, the complete five-source model achieved a lower mean ROC-AUC of 0.747 ± 0.117 .

Feature-selection order also influenced performance. Elastic Net followed by mRMR ranked first in 20 of the 31 source configurations, while mRMR followed by Elastic Net ranked first in 11. Correlation filtering followed by mRMR did not produce a configuration-level winner.

The overall effect of SMOTE was small. Although SMOTE produced a higher result in more configurations, most differences were less than 0.01 ROC-AUC, and the best overall pipeline did not use SMOTE.

Classifier performance also depended on the source configuration. RBF SVM was selected most frequently, followed by Logistic Regression and linear SVM. However, the highest overall performance was obtained with linear SVM.

The best model achieved a pooled ROC-AUC of 0.799, pooled PR-AUC of 0.832, balanced accuracy of 0.713, and F1-score of 0.746. It selected an average of 9.6 features per outer fold, although the exact selected features varied across folds.

Overall, the findings suggest that selective multi-source integration was more effective than complete feature concatenation and that model performance depended on the combined effects of source selection, feature selection, SMOTE, and classifier choice.

4.2 Effect of Multi-Source Feature Integration

Combining feature sources improved performance in some cases, but the improvement depended on which sources were combined.

The best mean ROC-AUC increased from 0.790 for a single-source dataset to 0.799 for a pairwise combination and 0.836 for a triple-source combination. However, performance decreased to 0.802 for the best quadruple-source configuration and

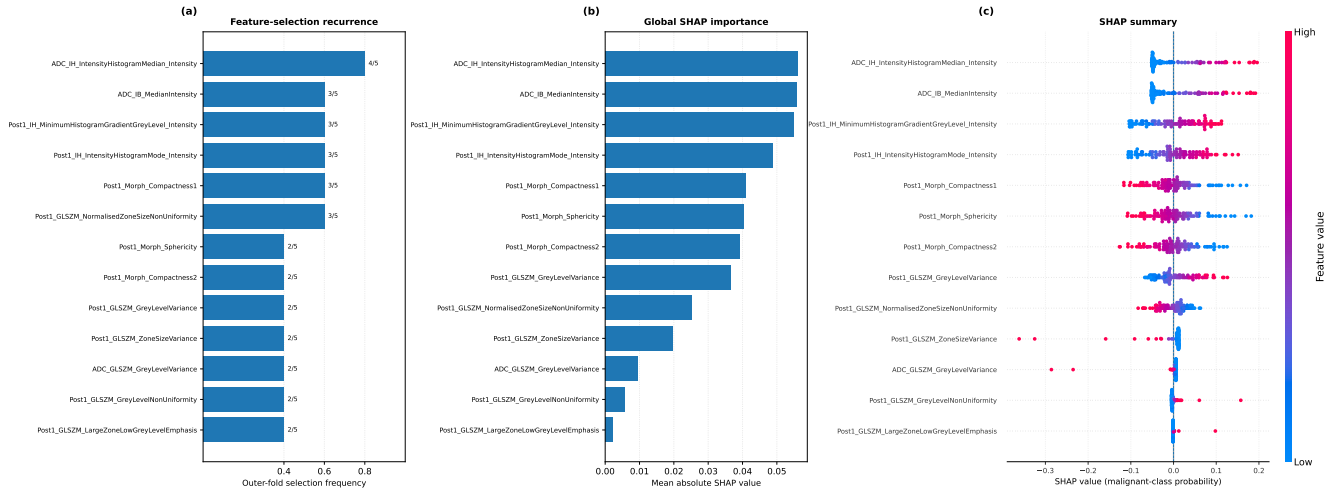


Figure 7: Feature-selection stability and SHAP interpretation of the highest-ranked ADC-Post1-Pre pipeline. (a) Selection frequency of features retained in at least two of the five outer folds. (b) Global feature importance measured by mean absolute SHAP value. (c) SHAP summary plot showing the magnitude and direction of feature contributions across ROIs. Positive SHAP values move the model output toward the malignant class, while negative values move it toward the benign class. Feature names were shortened for readability. P1, Post1; IH, intensity histogram; IB, intensity based; Comp., compactness; GL, grey level; ZS, zone size; ZSNU, zone-size non-uniformity; mRMR, minimum redundancy maximum relevance; SHAP, Shapley Additive Explanations; SVM, support vector machine. The full-cohort model was used only for interpretation; predictive performance was estimated from patient-aware out-of-fold predictions.

0.747 when all five sources were combined.

One possible explanation is the rapid increase in dimensionality. Single-source datasets contained about 145 predictors, compared with approximately 435 for triple-source datasets, 580 for quadruple-source datasets, and 724 for the complete five-source dataset. In contrast, the number of independent patients remained approximately 48.

Therefore, adding sources increased the number of candidate predictors without increasing the amount of independent training data. This can make feature selection more difficult and increase sensitivity to the patients included in each training fold.

The results also showed that the identity of the combined sources was more important than the number of sources alone. Different pairwise and triple-source combinations produced substantially different results despite having similar numbers of features and observations.

ADC appeared in many of the highest-ranked combinations. However, this does not prove that ADC alone was responsible for the improvement, because performance depended on its combination with other sources, the selected features, and the classifier.

These results support evaluating different source combinations directly rather than assuming that using all available sources will produce the best model.

4.3 Influence of Feature-Selection Strategy

Feature-selection strategy had a clear effect on model performance. Elastic Net followed by mRMR produced the highest result in 20 of the 31 source configurations, while mRMR followed by Elastic Net ranked first in the remaining 11.

Although these two strategies used the same methods, changing their order changed the intermediate candidate feature space and therefore the final selected subset.

In the Elastic-Net–mRMR pipeline, Elastic Net first performed a supervised and regularised re-

duction of the full feature space. mRMR then selected a smaller subset while considering both relevance and redundancy. This ordering produced the strongest average performance in the present study.

The reverse order remained competitive, showing that no single feature-selection sequence was optimal for every source configuration.

Correlation filtering followed by mRMR performed less favourably. Removing highly correlated predictors before supervised multivariable selection may remove variables that could still provide useful information when combined with other features.

Overall, the results show that feature selection should be considered part of the complete modelling pipeline rather than as an independent pre-processing step.

4.4 Effect of SMOTE and Class Imbalance

SMOTE did not consistently improve classification performance. It produced a higher mean ROC-AUC in 19 of the 31 source configurations, while no-SMOTE performed better in 12. However, the average difference between the two conditions was only about 0.003 ROC-AUC.

For 26 of the 31 configurations, the absolute difference was below 0.01. This indicates that the overall effect of SMOTE on discrimination was small.

The limited effect may partly reflect the relatively mild class imbalance in the datasets. For example, the complete five-source dataset contained 50 benign and 63 malignant ROIs.

SMOTE can still change threshold-dependent measures because it modifies the training distribution and therefore the fitted decision boundary. This may improve balanced accuracy, specificity, or precision even when ROC-AUC changes very little.

The two highest-ranked triple-source models and the complete five-source winner were obtained without SMOTE. In contrast, SMOTE was selected for several other successful pipelines, in-

cluding all five quadruple-source winners.

These results suggest that SMOTE should be tested as an optional part of the modelling pipeline rather than applied automatically.

4.5 Classifier Performance and Model Complexity

No classifier was optimal for every source configuration. RBF SVM was selected most often, appearing in 11 of the 31 highest-ranked pipelines. Logistic Regression was selected seven times and linear SVM six times.

The strongest overall model nevertheless used a linear SVM. This shows that the classifier selected most frequently was not necessarily the classifier producing the highest individual result.

The strong performance of linear SVM and Logistic Regression suggests that relatively simple regularised models remained competitive after feature selection. The selected feature subsets may have provided enough separation for linear decision boundaries to perform well.

Tree-based ensemble models were less frequently selected. XGBoost and LightGBM each appeared once, while Random Forest, Extra Trees, and Gradient Boosting did not produce any configuration-level winner.

This should not be interpreted as evidence that tree-based methods are generally unsuitable. Their performance may have been affected by the small number of independent patients and the limited hyperparameter search budget.

Each Optuna study used only ten trials. This provided a consistent comparison across the large experimental grid but may have been more restrictive for models with larger hyperparameter spaces.

Overall, classifier choice depended strongly on the source representation and selected features, with regularised linear and kernel-based models providing the most frequent high-performing solutions.

4.6 Feature Stability and Model Interpretation

The feature-stability analysis showed that strong predictive performance did not require exactly the same features to be selected in every training fold.

The best ADC–Post1–Pre pipeline selected an average of 9.6 features per outer fold. Across all five folds, 28 different features were selected at least once, 13 appeared in at least two folds, and none was selected in every fold.

This variability may result from the high-dimensional feature space and the relatively small number of independent patients. Several correlated or similarly informative predictors may provide comparable information, allowing different feature subsets to support similar classification performance.

The recurrent feature set contained features from ADC and Post1. No Pre-derived feature appeared in at least two folds. However, this does not mean that Pre was uninformative, because the best-performing source configuration included Pre and different Pre features may have contributed in individual folds.

SHAP analysis showed that the interpretation model relied most strongly on two ADC intensity features, together with several Post1 intensity, morphological, and texture features.

Feature-selection frequency and SHAP importance should be interpreted differently. Selection frequency measures how consistently a feature is retained across training samples, whereas SHAP describes its influence within a fitted model.

The SHAP model was refitted using the complete cohort only for interpretation. Therefore, the SHAP results describe the behaviour of the fitted model and should not be considered independent evidence of feature importance or biological causality.

4.7 Strengths and Limitations

A major strength of this study was the use of patient-aware nested cross-validation. All ROIs from the same patient remained in the same fold, reducing the risk of patient-level data leakage.

Preprocessing, feature selection, Auto- k , SMOTE, and hyperparameter optimisation were also restricted to the appropriate training data.

Another strength was the systematic evaluation of all 31 source combinations using the same modelling framework. This allowed single-source and multi-source representations to be compared under consistent conditions.

Several performance measures were also reported in addition to the primary mean outer-fold ROC-AUC, including pooled ROC-AUC, PR-AUC, threshold-dependent metrics, calibration, feature stability, and SHAP interpretation.

The main limitation was the small number of independent patients. Although more than 100 ROIs were available, the datasets represented approximately 48 patients. The effective sample size was therefore much smaller than the number of observations.

The study also did not include an independent external test cohort. Nested cross-validation reduces optimisation bias, but the best model was still selected from many candidate pipelines evaluated within the same cohort. External validation is therefore required.

Hyperparameter optimisation was limited to ten Optuna trials per study. This was necessary to keep the large computational experiment manageable but did not provide an exhaustive search, particularly for more complex classifiers.

Another limitation was the use of direct feature concatenation for multi-source integration. More advanced fusion methods may produce different results.

Finally, the selected features were not highly stable across outer folds, and SHAP interpretation was based on a separate full-cohort refit. These results should therefore be viewed as model explanations rather than evidence of a fixed radiomic signature or causal relationship.

Overall, the present framework provides a controlled comparison of candidate modelling strategies, but the results require confirmation in independent data.

4.8 Future Work and Generalisability

The most important next step is external validation of the ADC–Post1–Pre pipeline in an independent cohort. The complete pipeline, including preprocessing, feature selection, Auto- k , and classifier fitting, should be evaluated rather than testing only the final classifier.

Larger datasets with more independent patients would also improve the precision of performance estimates and provide a stronger assessment of feature-selection stability.

Future studies should investigate whether selective source integration continues to outperform complete concatenation in larger cohorts. Alternative fusion approaches, such as source-specific feature selection, prediction-level fusion, or representation-learning methods, could also be compared with direct concatenation.

A larger hyperparameter-optimisation budget may be useful for the most promising classifiers identified in this study, particularly models with larger search spaces.

Future work could also optimise the classification threshold using training data only and evaluate the selected threshold on independent patients. This would allow sensitivity and specificity to be adjusted for a predefined application.

Feature stability and SHAP interpretation should also be repeated in external datasets to determine whether the same sources, feature families, and individual predictors remain important.

Overall, the ADC–Post1–Pre pipeline represents the strongest candidate identified in the present study. However, larger and independent cohorts are required before its predictive performance, selected features, and model interpretation can be considered generalisable.

5 Conclusion

This study compared all 31 possible combinations of five MRI-derived radiomic feature sources using a patient-aware nested cross-validation framework. The results showed that combining more

feature sources did not always improve classification performance.

The best result was obtained with the ADC–Post1–Pre combination using Elastic Net followed by mRMR, Auto- k , no SMOTE, and a linear SVM. This pipeline achieved a mean outer-fold ROC-AUC of 0.836 ± 0.088 and a pooled ROC-AUC of 0.799.

Performance also depended on the feature-selection strategy, SMOTE condition, and classifier, showing that these components should be evaluated together as part of the complete modelling pipeline. Feature-selection stability analysis further showed that good predictive performance could be achieved even when the exact selected features changed across patient folds.

Overall, selective multi-source integration provided better results than complete feature concatenation in this dataset. However, the selected ADC–Post1–Pre pipeline was developed using a limited cohort, and validation in larger independent datasets is required before its performance and selected features can be considered generalisable.